

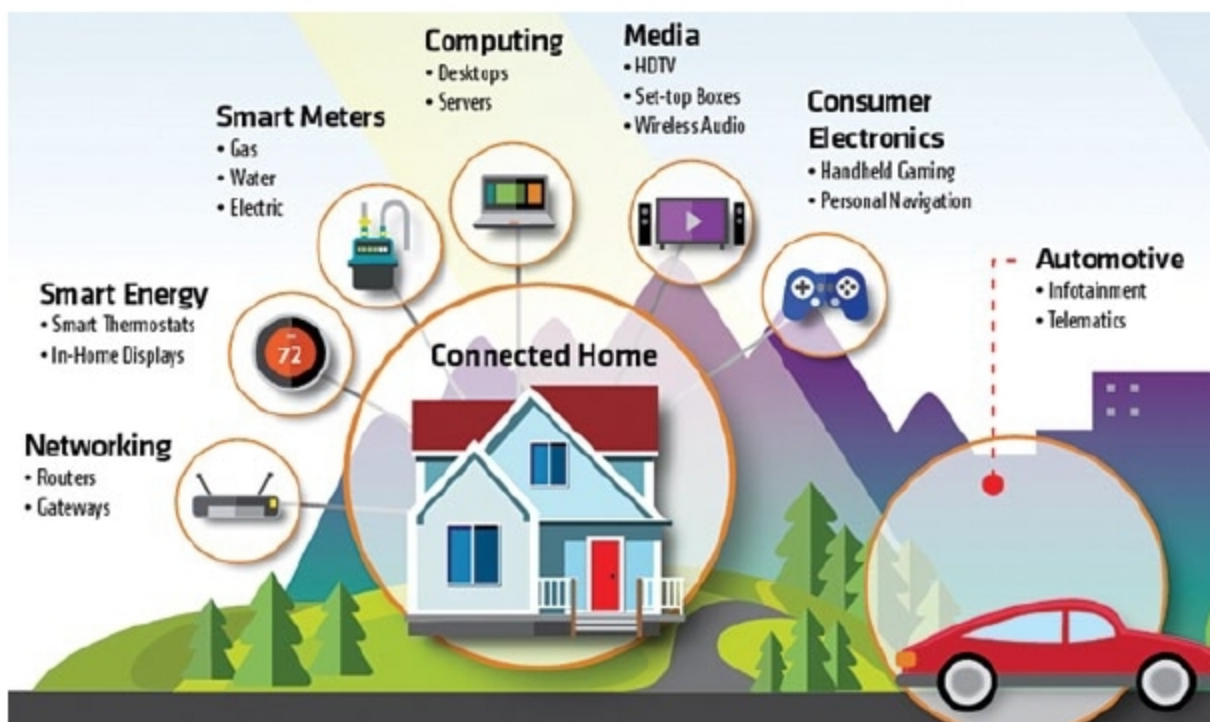


Fig. 4: Solutions for connected homes (Credit: Skyworks)

living.”

John Hamilton, vice president - marketing, RFMW Ltd, says, “We support components designed into IoT devices through Skyworks’ SKY85309-11 front-end module. It is a part of the RF section of an IoT radio. These products allow home-owners greater control over their home environment.

“A host of devices utilise Wi-Fi (IEEE 802.11), Zigbee 802.15.4 and Bluetooth products in smart thermostats, smoke and CO alarms, security system as well as routers, sensors, appliances and door locks, light switches, gaming consoles and televisions.”

Protocols and security features for wireless technology

Comba LTE Small Cell solution follows IPSEC 3GPP TS 36 104 security standards and protocols. Whereas, LoRaWAN follows AES 128 security standards and protocols.

EZVIZ applies the following secu-

rity standards and protocols for secure control:

1. ISO/IEC 27001 certified information security
2. Bank-level encryption security
3. Encrypting videos and images using user-defined passwords
4. Double-encrypting data before saving to cloud servers
5. Disabled SSH, Telnet and Web Service

6. EZVIZ privacy protocols
Bluetooth Mesh networking is designed with security as its number one priority. It is essential that the software in a connected device is not tampered with. Secure Boot feature provides surety that the firmware is authentic, has not been modified and cannot be downgraded.

u-blox security features are given below.

1. 3GPP standardised end-to-end security
2. Full two-way manageable channel

3. Security based on end-to-end 3GPP encryption

4. Data from device to cloud remains unchanged

5. Guaranteed quality of service (immunity to interference)

6. Device interoperability enabled by open standard protocols

7. Facilitate operations with horizontal IoT platforms

8. Embedded anti-tampering, life-time security services

9. Integrated secure element

There are many technology platforms and protocols for building a smart home. A smart home protocol must support a large number of devices, as well as offer the best possible device interoperability. Other factors are power consumption, bandwidth and, of course, cost.

Future scope of wireless technologies

The IoT connects things together through the cloud. Home automation is impossible if applications and systems are unable to connect with the network. A complete wireless network is an indispensable element of home automation. Thus, scope of home automation is not only about technologies for the applications, but also technologies for the networking system.

Enhancements in Mesh technologies improve performance and capacity. We are seeing an evolution of end-to-end security in constrained networks. Artificial intelligence (AI) used for human behaviours and patterns monitoring will result in an exponential growth of cellular and short-range technologies in the ecosystem of home automation. **EFY**

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